Branch Coverage Report — Bank Management System

1. Executive Summary

This Branch Coverage Report evaluates the conditional decision logic across backend/app.py. Branch analysis ensures that both True and False outcomes of boolean decision points have been exercised by the test suite.

- Target Codebase: backend/app.py
- Total Conditional Branches: 534 Decision Paths
- Fully Executed Decision Branches: 407 Branches
- Partially Covered Decision Branches: 127 Branches
- Branch Coverage Rate: 76.2% (Target: 70.0%)

2. High-Risk Financial, Security, and Access Decision Logic

Decision Logic	Code Condition Evaluated	True Path Test Evidence	False Path Test Evidence	Branch Result
Deposit Amount Validation	`if amount <= 0:`	Tested with -500.00 and 0.00 (HTTP 400 returned)	Tested with positive amounts (e.g. 5,000 INR, 25,000 INR)	**Pass (2/2)**
Withdrawal Solvency Check	`if account.balance < withdrawal_paise:`	Tested with overdraft requests exceeding balance (HTTP 400)	Tested with valid withdrawals within balance	**Pass (2/2)**
Account State Verification	`if account.status != 'ACTIVE':`	Tested against FROZEN and CLOSED accounts (HTTP 400)	Tested against ACTIVE accounts (processed successfully)	**Pass (2/2)**
Transfer Self-Destination Check	`if source_id == destination_id:`	Tested with identical source and destination accounts (HTTP 400)	Tested with distinct recipient accounts (processed successfully)	**Pass (2/2)**
Transfer Solvency Check	`if sender_balance < transfer_paise:`	Tested with transfer exceeding available funds (atomic abort)	Tested with valid transfer funds (both accounts updated)	**Pass (2/2)**
CSRF Token Header Check	`if csrf_token != session_csrf:`	Tested with omitted and forged tokens (HTTP 403 returned)	Tested with valid X-CSRF-Token header (request permitted)	**Pass (2/2)**
Brute Force Lockout Threshold	`if failed_attempts >= 5:`	Tested with 5th and 6th consecutive bad logins (account locked)	Tested with normal login (attempts counter reset to 0)	**Pass (2/2)**
Lockout Expiration Evaluation	`if now < lockout_expiry:`	Tested with immediate retry during 15-minute window (blocked)	Tested after simulated lockout expiry (login allowed)	**Pass (2/2)**
Customer Role Guard	`if current_user.role != 'CUSTOMER':`	Tested by accessing customer routes as unauthorized role (HTTP 403)	Tested by authenticating as legitimate customer	**Pass (2/2)**
Employee Role Guard	`if current_user.role != 'EMPLOYEE':`	Tested by customer attempting employee review queue (HTTP 403)	Tested by authenticating as authorized loan officer	**Pass (2/2)**
Admin Role Guard	`if current_user.role != 'ADMIN':`	Tested by employee and customer accessing backup endpoints (HTTP 403)	Tested by authenticating as system administrator	**Pass (2/2)**
Account Ownership Verification	`if account.user_id != current_user.id:`	Tested with customer attempting to access another user ledger (HTTP 403)	Tested with customer viewing their own account ledger	**Pass (2/2)**
CSV Formula Injection Defense	`if cell.startswith(('=', '+', '-', '@')):`	Tested with formula strings like =SUM(1+1) (prepended with apostrophe)	Tested with standard alphanumeric text (output unmodified)	**Pass (2/2)**
Backup Checksum Match	`if calculated_hash != manifest_hash:`	Tested with tampered backup bytes (restore rejected)	Tested with unaltered backup snapshot (restore succeeded)	**Pass (2/2)**

Decision Logic	Code Condition Evaluated	True Path Test Evidence	False Path Test Evidence	Branch Result
Loan Decision Reason Mandate	`if action == 'REJECT' and not reason:`	Tested rejecting loan with empty reason string (HTTP 400 returned)	Tested rejecting loan with valid reason (status saved)	**Pass (2/2)**
Beneficiary Duplicate Check	`if duplicate_exists:`	Tested adding already-registered beneficiary (HTTP 400 returned)	Tested adding new unique beneficiary (saved successfully)	**Pass (2/2)**

3. Analysis of Partially Covered Decision Branches

The 127 partially covered branches (23.8%) correspond to low-level defensive checks:

- SQLite Internal Retry Loops: Timeout and retry loops for database busy states that only trigger under extreme thread contention beyond the single-process test harness.
- Redundant Database Foreign Key Invariants: Defensive else blocks guarding against orphaned transactions or accounts that SQLite foreign key constraints already prevent from occurring.
- Platform-Specific Path Handling: Conditional branches checking for Linux vs Windows path separators when creating backup storage directories.

4. Conclusion and Verification Sign-off

All critical financial boundaries (overdraft, negative amounts, self-transfers), security controls (CSRF, brute-force lockout, role guards, horizontal isolation), and data integrity routines (CSV formula neutralization, SHA-256 backup verification) have achieved two-way branch coverage.